MILITARY SPECIFICATION

ANTENNA, FIXED, HIGH FREQUENCY, HORIZONTAL OMNIDIRECTIONAL BROADBAND

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the antenna described herein shall consist of this specification and the latest issue of MIL-A-28768.

1. SCOPE

1.1 Scope. This specification covers requirements for a fixed, omnidirectional, high frequency, horizontally polarized, broadband antenna, hereinafter referred to as the antenna.

1.2 Classification. Antennas shall be of the following types and classes, as specified (see 6.2):

Type	Frequency range
I	4.0 megahertz (MHz) to 30 MHz
II	6.0 MHz to 30 MHz
Class	
4A	120 miles per hour (mph), no ice and 75 mph with 1/4 inch (in.) of radial ice on all members
4B	140 mph, no ice and 60 mph with 1/2 in. of radial ice on all members
4C	140 mph, no ice and 90 mph with 1/2 in. of radial ice on all members

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specification and standard of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-28768 Antennas, Fixed, High Frequency, General Specification For

STANDARD

MILITARY

MIL-STD-1364 Standard General Purpose Electronic Test Equipment

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The antenna shall comply with MIL-A-28768 and the requirements specified herein. The antenna shall be furnished as a complete system ready for installation and operation. The antenna shall be designed for broadband operation without tuning or adjustment.

3.2 Size. The antenna shall comply with the size limitations specified in TABLE I.

TABLE I. Size of antennas.

	Type I (m) 1/	Type I (ft) 2/	Type II (m)	Type II (ft)
Maximum length, width, or diameter, including guys and anchors	88.4	290	51.9	190
Maximum height	73.2	240	51.8	170

1/ meters
2/ feet

3.3 Environmental wind and ice conditions. The antenna shall be designed to conform to the allowable stress requirements of MIL-A-28768 when loaded with the wind and ice conditions as specified by the Class (see 1.2). Unless otherwise specified (see 6.2), Class 4A shall apply. Structural relief devices (fuses) shall not be used to conform to the environmental conditions.

3.3.1 Snow and ice design. The antenna shall be designed such that snow and ice will not accumulate at the feedpoint and inside the apex at the base of the antenna to the extent that allowable stress requirements are exceeded or performance requirements are degraded below those specified herein.

3.4 Structural requirements. Structural requirements shall be as specified in 3.4.1 through 3.4.3.

3.4.1 Rotational stability. The antenna shall be designed with an appropriate guying or bracing system which shall eliminate the possibility of rotational instability. The antenna shall be designed so that rotation about a vertical axis through the center of the antenna shall be not greater than 30 minutes of arc.

3.4.2 Field adjustment of cable tension. The design of the antenna shall provide facilities for field adjustment of cable tensions both during and after erection.

3.4.3 Installation design. The antenna shall be designed to be installed on sites with conditions ranging from flat and level to slopes of up to 3 degress in any direction. All additional parts and materials such as guy material, anchors, and attachment hardware which are necessary to permit antenna installation over the site conditions specified herein shall be supplied. This requirement shall be achieved with all towers remaining vertical (plumb).

3.5 Electrical performance. Electrical performance requirements shall be as specified in 3.5.1 through 3.5.7.

3.5.1 Frequency range. The antenna shall operate over the entire frequency range as specified by the Type (see 1.2). Unless otherwise specified (see 6.2), Type I shall apply.

3.5.2 Power. Unless otherwise specified (see 6.2), the antenna shall be capable of the power level specified by Power 25 in MIL-A-28768.

3.5.3 Gain. The antenna shall have the minimum gain specified in TABLE II for the corresponding frequencies.

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TABLE II. Pattern and gain.

Frequency (MHz)	Main beam (minimum) (degrees)	Main beam (maximum) (degrees)	UHPP 1/ (minimum) (degrees)	LHPP 2/ (maximum) (degrees)	Gain (minimum) (dBi) 3/
4	29	31	50	15	6.5
8	19	23	30	10	7.5
12	16	19	25	8	7.0
16	13	16	22	7	7.0
30	8	10	15	5	7.0

1/ Upper half power point (see 6.3)
2/ Lower half power point (see 6.3)
3/ Decibels isotropic

3.5.4 Polarization. The antenna shall be horizontally or elliptically polarized.

3.5.5 Radiation pattern. The antenna shall have a radiation pattern as specified in 3.5.5.1 and 3.5.5.2.

3.5.5.1 Azimuth. The antenna shall be omnidirectional within ±1 decibel.

3.5.5.2 Elevation. The location of the elevation beam's maximum gain point, the UHPP, and LHPP shall be within the ranges specified in TABLE II for the corresponding frequencies. The points are specified in degree of angle.

3.5.6 Input impedance. The nominal input impedance shall be 50 ohms.

3.5.6.1 Voltage standing wave ratio (VSWR). The maximum VSWR shall be 2.5 to 1 with respect to 50 ohms.

3.5.7 Efficiency. The efficiency of the antenna, when installed over perfectly conducting ground, shall be not less than 90 percent at a VSWR of 1 to 1.

3.6 Ground system. The antenna shall not be dependent upon an artificial ground system to obtain the performance specified herein.

3.7 General purpose electronic test equipment (GPETE). The antenna shall be designed so that all testing can be accomplished using GPETE selected from MIL-STD-1364. Any nonstandard and special purpose electronic test equipment, if necessary for antenna testing, shall be subject to the approval of the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance provisions shall be in accordance with MIL-A-28768 and as specified in 4.2.

4.2 First article inspection. In addition to the first article inspections specified in MIL-A-28768, the inspections in a through d shall be conducted:

a. Verification of size requirements of 3.2
b. Verification of rotational stability requirements of 3.4.1
c. Verification of installation design requirements of 3.4.3
d. Verification of VSWR requirements of 3.5.6.1

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preparation for delivery shall be as specified in MIL-A-28768.

6. NOTES

6.1 Intended use. The antenna will be used for general purpose radio communications and is intended for permanent installation at shore locations.

6.2 Ordering data. In addition to the items specified in MIL-A-28768, procurement documents should specify:

a. Title, number, and date of this specification
b. Type, if other than Type I is required (see 1.2 and 3.5.1)
c. Class, if other than Class 4A is required (see 1.2 and 3.3)
d. Power, if other than Power 25 as specified in MIL-A-28768 is required (see 3.5.2)

6.3 Half power points. The UHPP and the LHPP are the locations on the radiation pattern where the radiated power is one-half that of the main beam, above (upper) and below (lower) the line which connects the beam maximum to the center of the polar graph of the radiation pattern.

Preparing activity:
NAVY-EC

(Project No. 5985-N515)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28768/4(EC)	ANTENNA, FIXED, HIGH FREQUENCY, HORIZONTAL OMNIDIRECTIONAL BROADBAND

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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